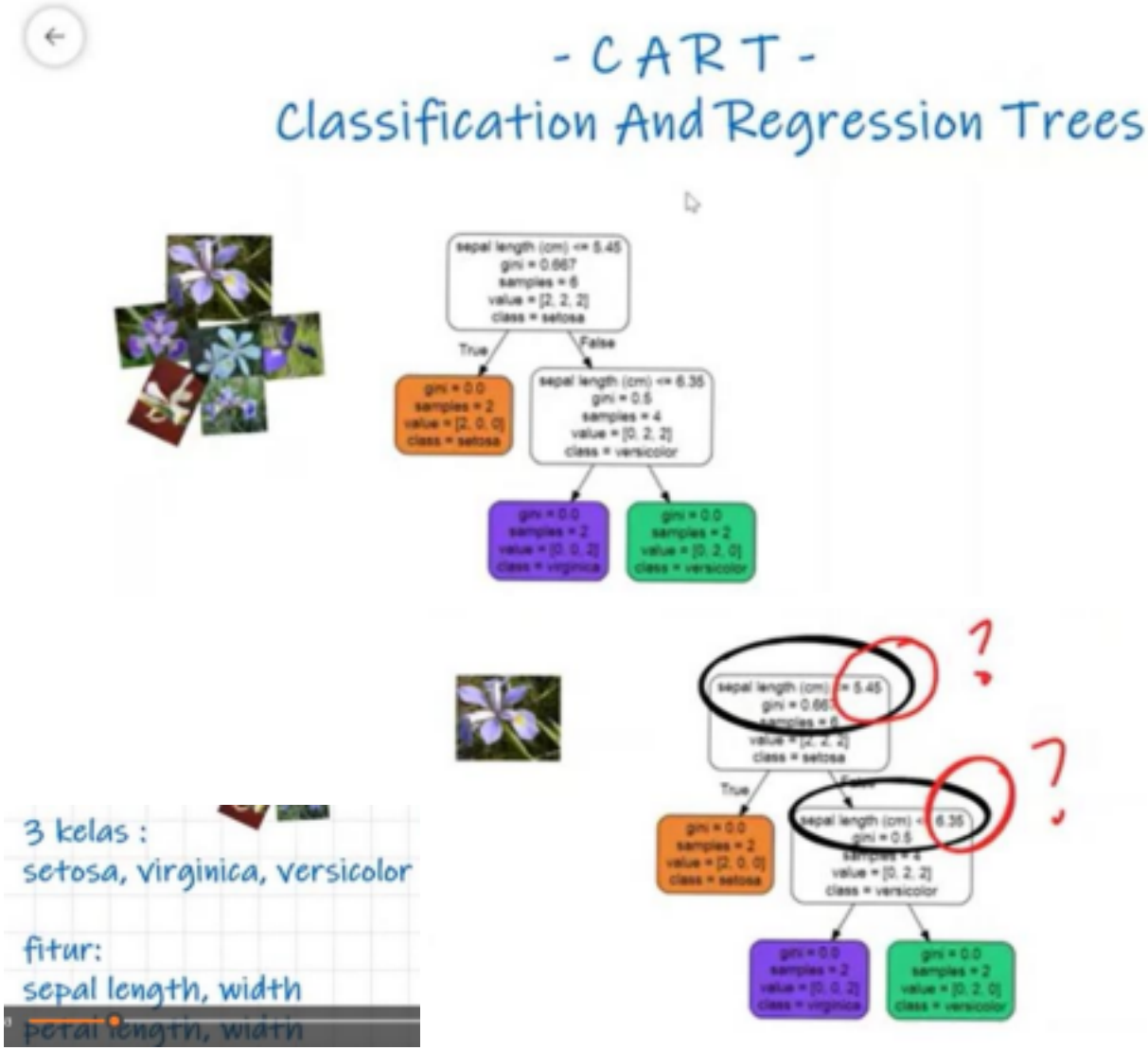
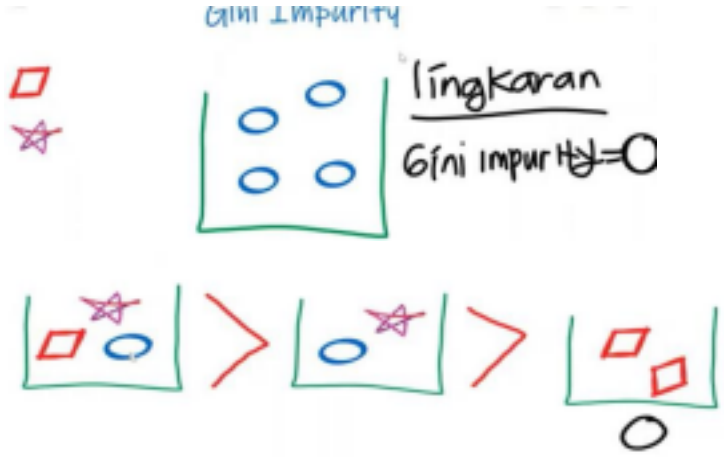


Gini impurity: berapa % kesalahan dalam menentukan klasifikasi



GiniImpurity=

$$\sum_{i=1}^{c} P_i \cdot (1-P_i)$$

1 Kelas

$P_1 = \frac{2}{2} = 1$

GiniImpurity =

$P_1 \cdot (1-P_1) = 1 \cdot (1-0)$

$= 0$ //

2 Kelas

$P_1 = \frac{1}{2}$ $P_2 = \frac{1}{2}$

0 //

Gini

P_1

Gini $= P_1(1-P_1) + P_2 \cdot (1-P_2)$

$= \frac{1}{2}(1-\frac{1}{2}) + \frac{1}{2}(1-\frac{1}{2}) = \frac{1}{2}$

npurity=

3 Kelas

$P_1 = \frac{1}{3}$

$P_2 = \frac{1}{3}$

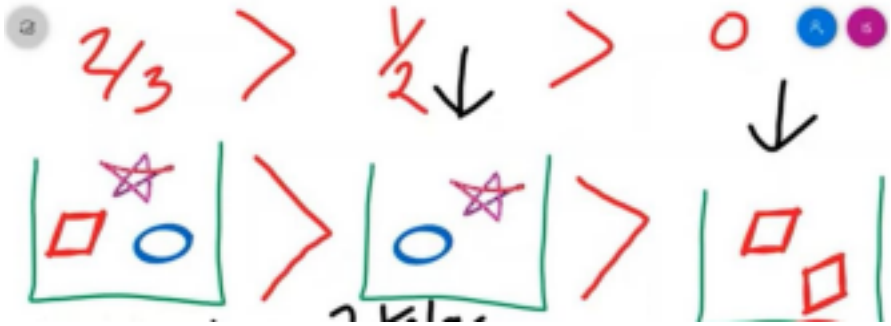
$P_3 = \frac{1}{3}$

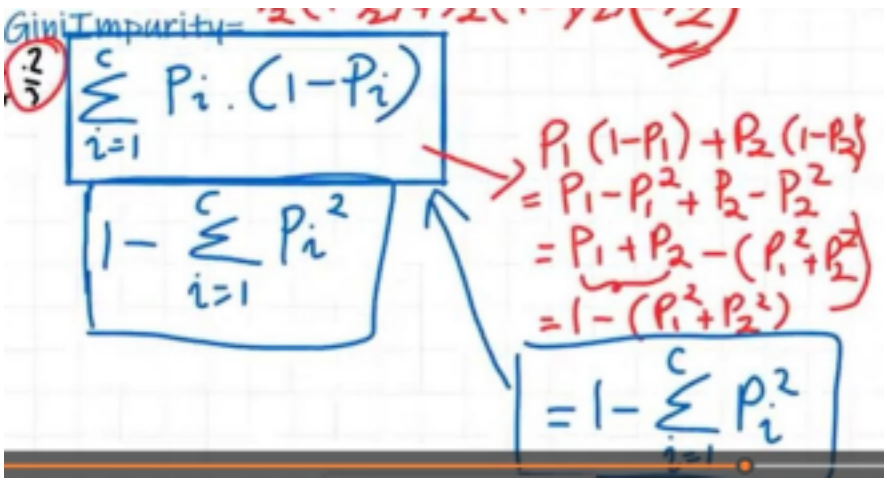
$P_1(1-P_1) + P_2(1-P_2) + P_3(1-P_3)$

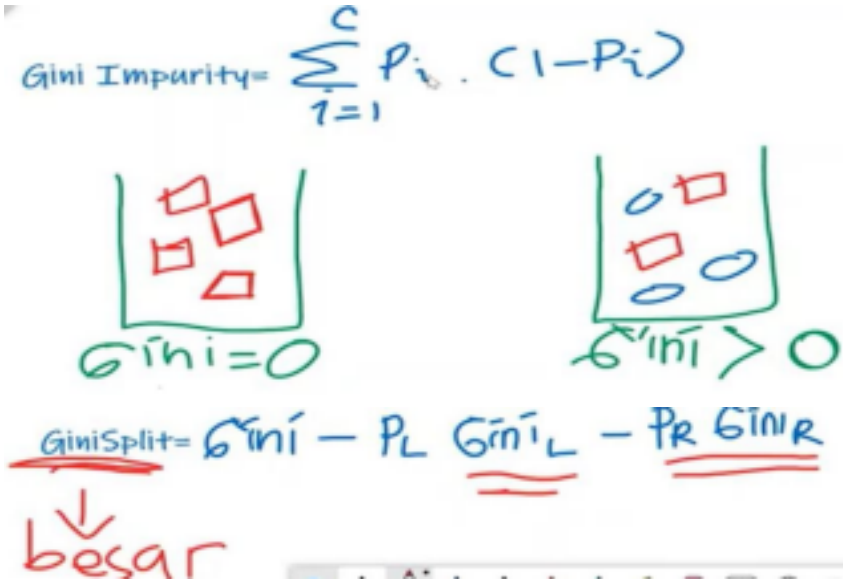
$= \frac{1}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{2}{3} +$

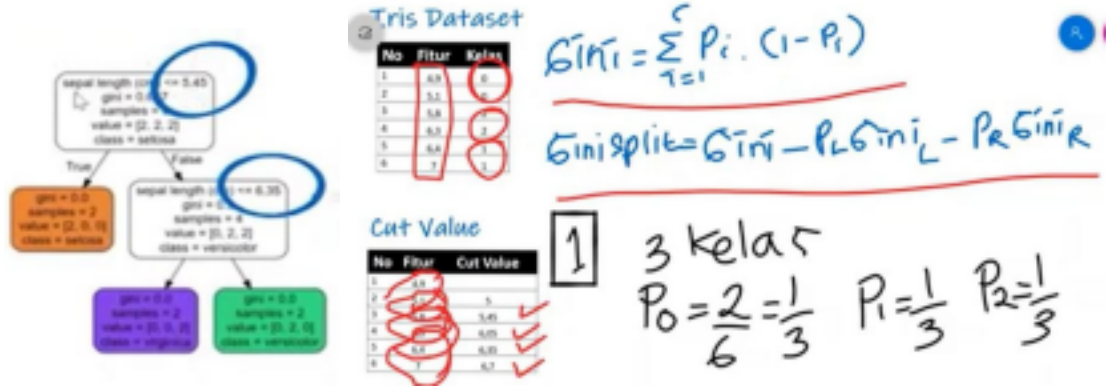
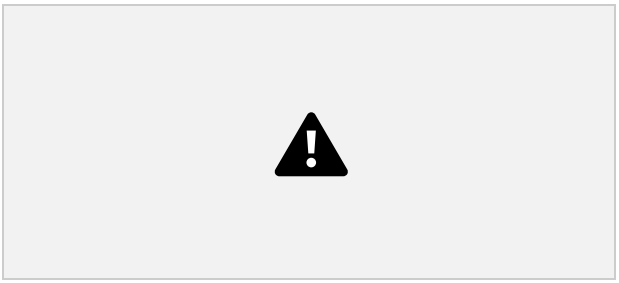
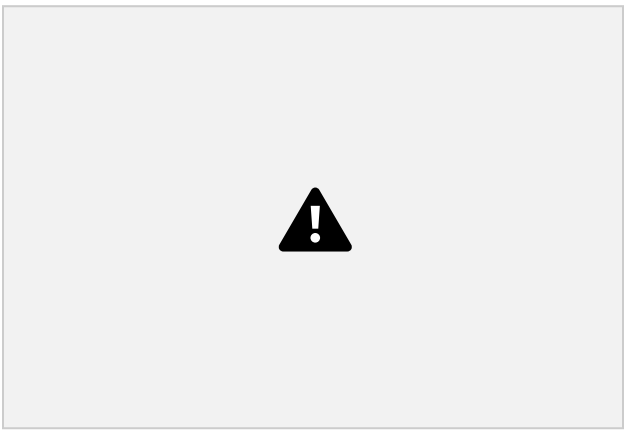
$\frac{1}{3} \cdot \frac{2}{3} = \frac{1}{3} \cdot \frac{2}{3}$

GiniIm


2/3 > 1/2 > 0
2 kelas


GiniImpurity=
2/3
$\sum_{i=1}^{c} P_i . (1-P_i)$
$1-\sum_{i=1}^{c} P_i^2$
$P_1(1-P_1)+P_2(1-P_2)$
$= P_1-P_1^2+P_2-P_2^2$
$= P_1+P_2-(P_1^2+P_2^2)$
$=1-(P_1^2+P_2^2)$
$=1-\sum_{i=1}^{c} P_i^2$


Gini Impurity= $\sum_{i=1}^{c} P_i . (1-P_i)$
Gini=0
Gini > 0
GiniSplit= Gini − P_L $Gini_L$ − P_R $Gini_R$
besar


Iris Dataset
No Fitur Kelas
Cut Value
No Fitur Cut Value
$Gini = \sum_{i=1}^{c} P_i . (1-P_i)$
GiniSplit= Gini − P_L $Gini_L$ − P_R $Gini_R$
1
3 kelas
$P_0 = \frac{2}{6} = \frac{1}{3}$ $P_1 = \frac{1}{3}$ $P_2 = \frac{1}{3}$